

Reflect shapes

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Reflect $(-6, 2)$ in the y-axis. Copy or complete the first step.

2. A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.

3. Plot $A(-4, 3)$. Which quadrant contains A?

4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Reflect $(-6, 2)$ in the y -axis. Copy or complete the first step.
Answer $(6, 2)$.
2. A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.
Answer $(-2, -1)$.
3. Plot $A(-4, 3)$. Which quadrant contains A?
Answer Quadrant II.
4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.

Reflect shapes

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Reflect $(-6, 2)$ in the y-axis.

6. Check this result using an inverse, estimate or known fact: Plot $A(-4, 3)$. Which quadrant contains A?

2. A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.

7. Prove or explain your reasoning: Point P moves from $(2, -5)$ to $(-3, 1)$. Describe the translation.

3. Plot $A(-4, 3)$. Which quadrant contains A?

8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.

4. Solve this independently and show every stage: Reflect $(-6, 2)$ in the y-axis.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.

10. Write and solve a similar question to: Plot $A(-4, 3)$. Which quadrant contains A?

Teacher answers and checking prompts

1. Reflect $(-6, 2)$ in the y-axis.
Answer $(6, 2)$.
2. A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.
Answer $(-2, -1)$.
3. Plot A $(-4, 3)$. Which quadrant contains A?
Answer Quadrant II.
4. Solve this independently and show every stage: Reflect $(-6, 2)$ in the y-axis.
Answer $(6, 2)$.
5. Use a second method or representation for: A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.
Answer Expected result: $(-2, -1)$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Plot A $(-4, 3)$. Which quadrant contains A?
Answer Expected result: Quadrant II. A valid check must be shown.
7. Prove or explain your reasoning: Point P moves from $(2, -5)$ to $(-3, 1)$. Describe the translation.
Answer 5 left and 6 up. The explanation must show why the method works.
8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Plot A $(-4, 3)$. Which quadrant contains A?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reflect shapes

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Point P moves from (2, -5) to (-3, 1). Describe the translation.

6. Create a new example that tests the same idea as: Plot A(-4, 3). Which quadrant contains A?

2. Prove the answer to this question, rather than only calculating it: Reflect (-6, 2) in the y-axis.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: Quadrant II.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer 5 left and 6 up. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Reflect (-6, 2) in the y-axis.
Answer Expected result: (6, 2). Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
Answer Expected result: (-2, -1). Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: Quadrant II.
Answer One valid reconstruction is: Plot A(-4, 3). Which quadrant contains A? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
6. Create a new example that tests the same idea as: Plot A(-4, 3). Which quadrant contains A?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.